Lesson Planning

Complete Lesson Plan Book

Class: Class 7

Subject: Science

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Simple Machines: Levers, Pulleys and Inclined Planes

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Unit 1: Simple Machines: Levers, Pulleys and Inclined Planes

Learning Objectives

• Students will identify levers, pulleys and inclined planes in everyday objects.

• Students will explain how a simple machine makes work easier by changing force or direction.

• Students will compare a first-class lever, a fixed pulley and a ramp using examples.

Key Concepts

• A simple machine is a device that helps us do work with less effort, either by increasing the force applied or by changing its direction.

• A lever is a bar that turns around a fixed point called the fulcrum; a seesaw and a crowbar are levers.

• A fixed pulley changes the direction of a force: pulling down on a rope lifts a load up, as with a flagpole pulley.

• An inclined plane is a sloping surface, such as a ramp, that lets a heavy load be raised gradually instead of lifted straight up.

• Machines do not create energy; they trade a smaller force over a longer distance for a larger force over a shorter distance.

Teaching Methodology (Step by Step)

1. Ask students how they would lift a heavy box onto a truck, and collect ideas involving ramps, ropes and bars.

2. Define a simple machine and introduce the lever, pulley and inclined plane with a diagram of each.

3. Demonstrate a ruler lever lifting a book, a pulley model lifting a small weight, and a ramp rolling a ball up a slope.

4. Students sort pictures of everyday objects into lever, pulley and inclined-plane groups and justify each choice.

5. Students explain in pairs which machine they would choose for a given job and why.

Classroom Activities

• Lever hunt: students find three levers at home or school and name the fulcrum, load and effort in each.

• Pulley model: groups build a simple pulley with a spool and string and lift a pencil case with it.

• Ramp race: students roll a toy car up ramps of different slopes and record which slope needs the least push.

Assessment Questions

1. Name one lever, one pulley and one inclined plane found in a school.

2. How does a fixed pulley make lifting easier if it does not reduce the force?

3. Why is it easier to push a heavy box up a long gentle ramp than up a short steep one?

Homework

• Draw and label a lever, a pulley and an inclined plane from your home, naming the fulcrum or the slope in each.

• Write four lines explaining how a wheelbarrow uses a lever to carry a heavy load.

Time Allocation

40 minutes

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